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Classification of human hand movements based on EMG signals using nonlinear dimensionality reduction and data fusion techniques

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ABSTRACT

Surface electromyography (EMG) is non-invasive signal acquisition technique that plays a central role in many application, including clinical diagnostics, control for prosthetic devices and for human-machine interactions. The processing typically begins with a feature extraction step, which may be followed by the application of a dimensionality reduction technique. The obtained reduced features are input for a machine learning classifier. The constructed machine learning model may then classify new recorded movements.

The features extracted for EMG signals usually capture information both from the time and from the frequency domain. Short time Fourier transform (STFT) is commonly used for signal processing and in particular for EMG processing since it captures the temporal and the frequency characteristics of the data. Since the number of calculated STFT features is large, a common approach in signal processing and machine learning applications is to apply a linear or a nonlinear dimensionality reduction technique for simplifying the feature space. Another aspect that arises in medical applications in general and in EMG based hand classification in particular, is the large variability between subjects. Due to this variability, many studies focus on single subject classification. This requires acquiring a large training set for each tested participant which is not practical in real life application.

The objectives of this study were first to compare between the performances of a nonlinear dimensionality technique to a standard linear dimensionality method when applied for single subject EMG based hand movement classification, and to examined their performances in case of limited amount of training data samples. The second objective was to propose an algorithm for multi-subjects classification that utilized a data alignment step for overcoming the large variability between subjects.

The data set included EMG signals from 5 subjects who perform 6 different hand movements. STFT was calculated for feature extraction, principal component analysis (PCA) and diffusion maps (DM) were compared for dimension reductions. An affine transformation for aligning between the reduced feature spaces of two subjects, was investigated. K-nearest neighbors (KNN) was used for single and multi-subject classification.

The results of this study clearly show that the DM outperformed the PCA in case of limited training data. In addition, the multi-subject classification approach, which utilizes dimension reduction methods along with an alignment algorithm enable robust classification of a new subject based on another subjects' data sets. The proposed framework is general and can be adopted for many EMG classification task.

1. Introduction

Machine learning is integrated nowadays in many data driven application, such as signal and image processing tasks, due to its generic and flexible methodologies. The rapid development of advanced and sophisticated signal acquisition tools, which results with large amounts of complex data, gave rise to new

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branches in machine learning that deal with compact modeling of these data sets. Coupling dimensional reduction methods with time series processing is a natural match and recent work in diverse fields (Dov, Talmon, & Cohen, 2016; Dsilva, Talmon, Coifman, & Kevrekidis, 2018; Duncan, Talmon, Zaveri, & Coifman, 2013; Rabin, Bregman, Lindenbaum, Ben-Horin, & Averbuch, 2016; Talmon, Cohen, Gannot, & Coifman, 2013) justify this claim. Algorithms that utilize machine learning for signal processing typically include a number of consecutive steps. First, features are extracted from the time series signal. Next, when the dimension of the feature vector is large, a dimensionality reduction method is used to construct a compact representation of the dataset. Last, a supervised or an unsupervised method is applied to predict the class of a new recorded signal.

The use of electromyography (EMG) signal allows convenience and non-invasive access signal acquisition resulting in new developed methods for classification and feature extraction. Identification of hand movements based on EMG measurements have been largely used in the field of computer and automatic video games, robotic exoskeleton, operative devices and for power prostheses (Batzianoulis et al., 2017; Gailey, Artemiadis, & Santello, 2017; Na, Kim, Jo, & Kim, 2017) and thus, has been the subject of many studies over the past few years.

A large number of these studies focus in features selection for EMG movement classifications and included in their methods is a dimensionality reduction step followed by a machine learning based classifier. The dimensionality reduction step is typically performed by a linear dimensionality reduction technique such as Principal Component Analysis (PCA), which is limited in representing complex data that has in it nonlinear relationships. Nonlinear dimensionality techniques enable to preserve the local structures in the original feature space and by that overcome some of PCA's limitations, due to the method's global and linear nature. However, while the extension to new data points in the linear dimensionality reduction methods is straightforward, methods based on the Nyström technique (Coifman and Lafon, 2006b) are used for the extension of nonlinear models.

Selection of significant discriminative features for EMG hand movement classification is discussed in several recent papers. Phinyomark et al. (2013) computed 50 features from EMG data that was measured in a period of 21 days from a single male subject, to identify a feature set that is stable over a long period of time. Using a linear discriminate analysis classification algorithm it was found that the sample entropy was the most stable feature. In another study (Tsai, Luh, & Lin, 2015) it was found that the Short time Fourier transform (STFT) ranking feature gives higher classification rates (93.9%) compared with conventional EMG features as mean absolute value, zero crossing, slope sign change, waveform length, auto regression, median and mean spectrum frequency (33–90.8%) for motion pattern recognition from multi-channel EMG signals of 6 muscles. The linear PCA method was applied in this work for dimensionality reduction and the Support Vector Machines (SVM) for the classification. Other studies investigate the best features selection for pattern recognition classification of myoelectric limb prostheses (Al-Angari, Kanitz, Tarantino, & Cipriani, 2016; Khushaba, Al-Timemy, Kodagoda, & Nazarpour, 2016). Khushaba et al. (2016) studied the effect of both forearm orientation and muscle contraction on the classification accuracy using wavelet transform based features and time and frequency domain features. Al-Angari et al. (2016) investigate the best pairs of features and channels for the classification of 5 hand postures at 9 different arm positions. Using 10 features from the time and frequency domain two methods for feature-channels pairs selection were compared, distance based and correlation-based feature selection. In both studies the results were evaluated for each subject separately, based only on his/her train data. The performance of a convolutional neural network classifier was compared with two other classifiers (linear discriminant analysis and stacked sparse autoencoders) for the classification of eight hand motions that were repeatedly performed by each of seven subjects. Features from the time domain were computed and fed as inputs to the classifiers and the results, in line with the previous all above works were evaluated for each subject separately.

Several studies have also been conducted over the years to investigate the ability to identify hand movements using different time and frequency domain EMG features. These EMG features were used in a study aimed to identify six daily hand movements (Sapsanis, Georgoulas, & Tzes, 2013). Two dimensionality reduction techniques, PCA (Pearson, 1901) and RELIEF (Kira et al., 1992) were used in this study to reduce the dimension of the feature space. Then, the classification was carried out on each subject separately, where 50%–70% of the data were used as train points. The reduced input was then inserted into a linear classifier and resulted with an average of 85% success classification rate without significant difference between the two methods. The dataset used by Sapsanis et al. (2013), which is publicly available in the UCI repository (UCI datasets), was further tested in several recent studies (Gu, Zhang, Zhao, & Luo, 2017; Ramírez-Martínez, Alfaro-Ponce, Pogrebnyak, Aldape-Pérez, & Argüelles-Cruz, 2019; Ruangpaisarn & Jaiyen, 2015). Gu et al. (2017) used the above dataset to compare between several machines learning algorithms for classification after calculating featured with Empirical Mode Decomposition (EDM). The compared classification methods were Neural Network, Adaptive Boosting, Linear Discriminant Analysis, Random Forest and Random Forest with PCA. Using 80% of the data from all the subjects as the training set, the Neural Network classifier resulted in accuracy of 85% while the Adaptive Boosting and the Linear Discriminant Analysis achieved only a 55% and 65% correct classification rate, respectively. An increase in the accuracy rates to 91% and 94% was achieved with the Random Forest and Random Forest with PCA, correspondingly.

In Ruangpaisarn and Jaiyen (2015), where Singular Value Decomposition (SVD) was applied for feature extraction, Naive Bayes, Radial Basis Function Network, k-nearest neighbors (KNN) and SVM trained by Sequential minimal optimization (SMO) (Platt, 1999), were compared for classification. The classification step was performed on each subject separately achieving accuracies of 91.66%, 94%, 94.77% and 98.22%, correspondingly. More recently, Ramírez-Martínez et al. (2019) examined the use of Burg reflection coefficients and tested many combinations of different feature-based datasets with machine learning algorithms for classification of the UCI hands movement dataset. A ten-fold cross-validation procedure was applied to the full dataset that held all of the subject's data. Results for a specific setting of an instance based classifier reached the accuracy of 100%, using 60 features. The results degraded a bit when the number of features was reduced.

Several other public and not-public dataset were tested in several papers, with the task being gesture or hand movement classification. In (Tang, Liu, Lv, & Sun, 2012), the authors collected EMG signals from multiple channels in order to discriminate between eleven hand gestures. In this study the dimensionality reduction step was bypassed by extracting only a small number of informative EMG features such as energy ratios and concordance correlation coefficient. A cascaded-classifier that was based on 25 train movements of each type was applied on each subject separately to the feature set and resulted in an accuracy rate of 89%. A real-time EMG recognition algorithm for the control of multifunction myoelectric hand was developed based on wavelets for feature extraction (Chu, Moon, & Mun, 2006). PCA followed by Self Organization Maps and a multi-layer neural network was used for dimensional-

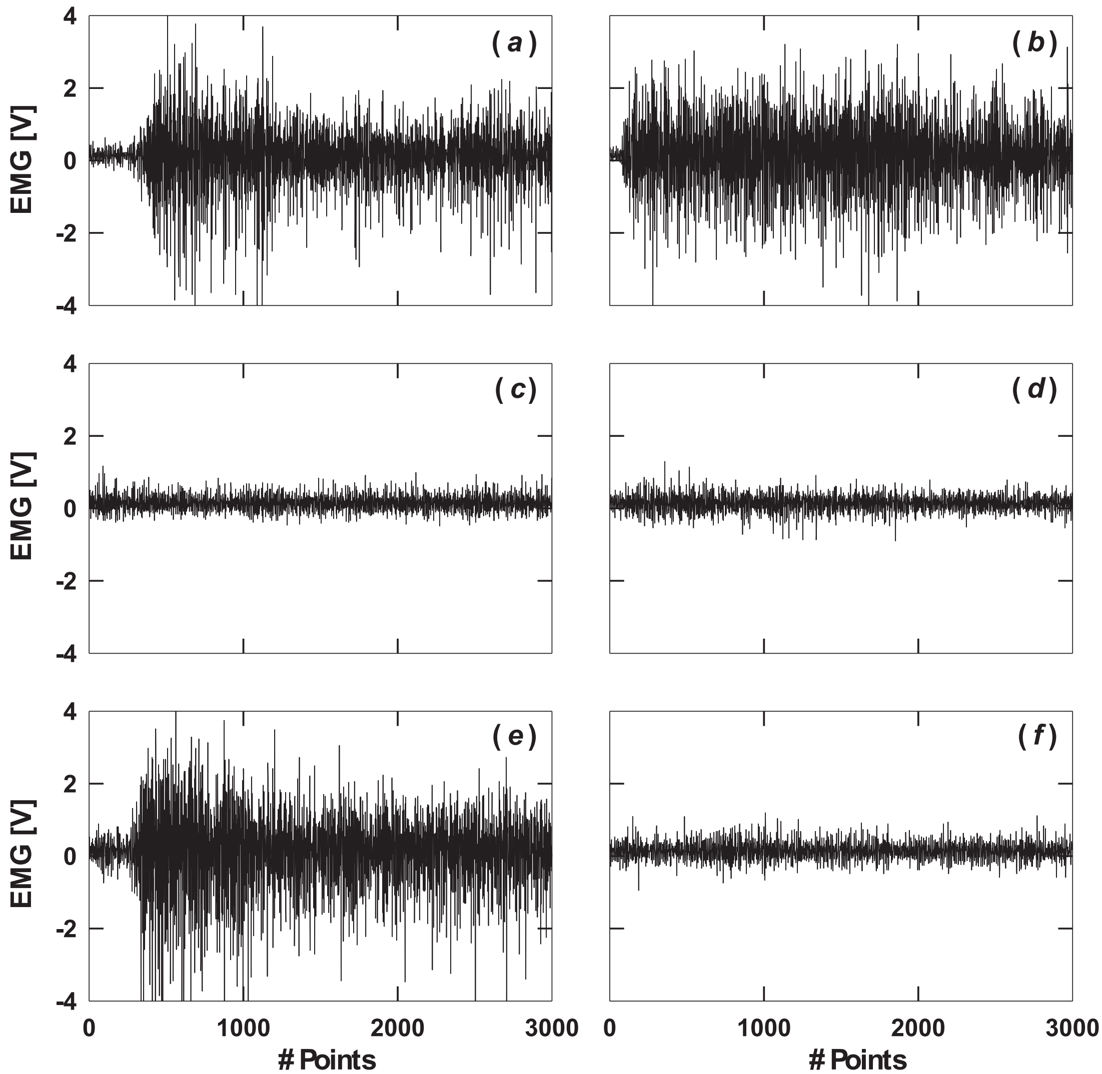

Fig. 1. An example of the rawdata of EMG signals during the six hand movements of subject #1. (a) Cylindrical, (b) Hook, (c) holding Tip, (d) Palmar, (e) Spherical and (f) Lateral.

ity reduction while using data from the same subject for the train and for the test. Discrete Wavelet feature along with Artificial Neural Network classifier was applied in Mane, Kambli, Kazi, & Singh, 2015, and was optimized to each of four subjects separately that performed three basic hand movements, with an overall correct classification rate of 93%. A PCA based algorithm was also used to drive an under actuated prosthetic hand prototype having a two dimensional control input (Matrone, Cipriani, Secco, Magenes, & Carrozza, 2010) and for classification of 52 hand movements using an individual classifier that was constructed for each subject (Isaković, Miljković, & Popović, 2014).

In all of the above previous studies the computed classification algorithms, as far as we know, were tested on each subject separately, and were usually based on a relatively large number of training samples. The presented classification results in these works represent the average correct classification rates from all of the subjects. In other words, data fusion for merging between several subjects, is not performed. In order to achieve high classification rates for a single subject classification algorithm, one is required to record a large amount of training examples. In real life application, this request is not practical. Constructing a fused multi-subject representation is challenging. The variance between the signals recorded from two subjects that perform the same set of hand movements is large due to the different physical characteristics of the subjects. Thus, a simple step of merging between training data of different subjects is bound to fail. Recently, deep-learning methods were applied for creating multi-subject models.

In Côté-Allard et al. (2019), a deep-learning algorithm, which uses transfer learning, was applied for automatic hand movement classification. Time and frequency information was computed in the features extraction step. Linear Discriminant Analysis was tested for dimensionality reduction. Two datasets, which comprise of 19 and 17 participants were used as input for the deep learning method. Transfer learning implemented the data fusion between different subjects.

Other deep learning based methods were proposed for multi-subject classification (Phinyomark & Scheme, 2018; Tsinganos, Cornelis, Cornelis, Jansen, & Skodras, 2018). These methods typically compute STFT features (similarly to what is proposed in this paper), and treat them as input images for Convolutional Neural Networks.

Although deep learning based models, and in particular those which include transfer learning, are suitable for fusing samples recorded from different subjects, large amounts of data is required

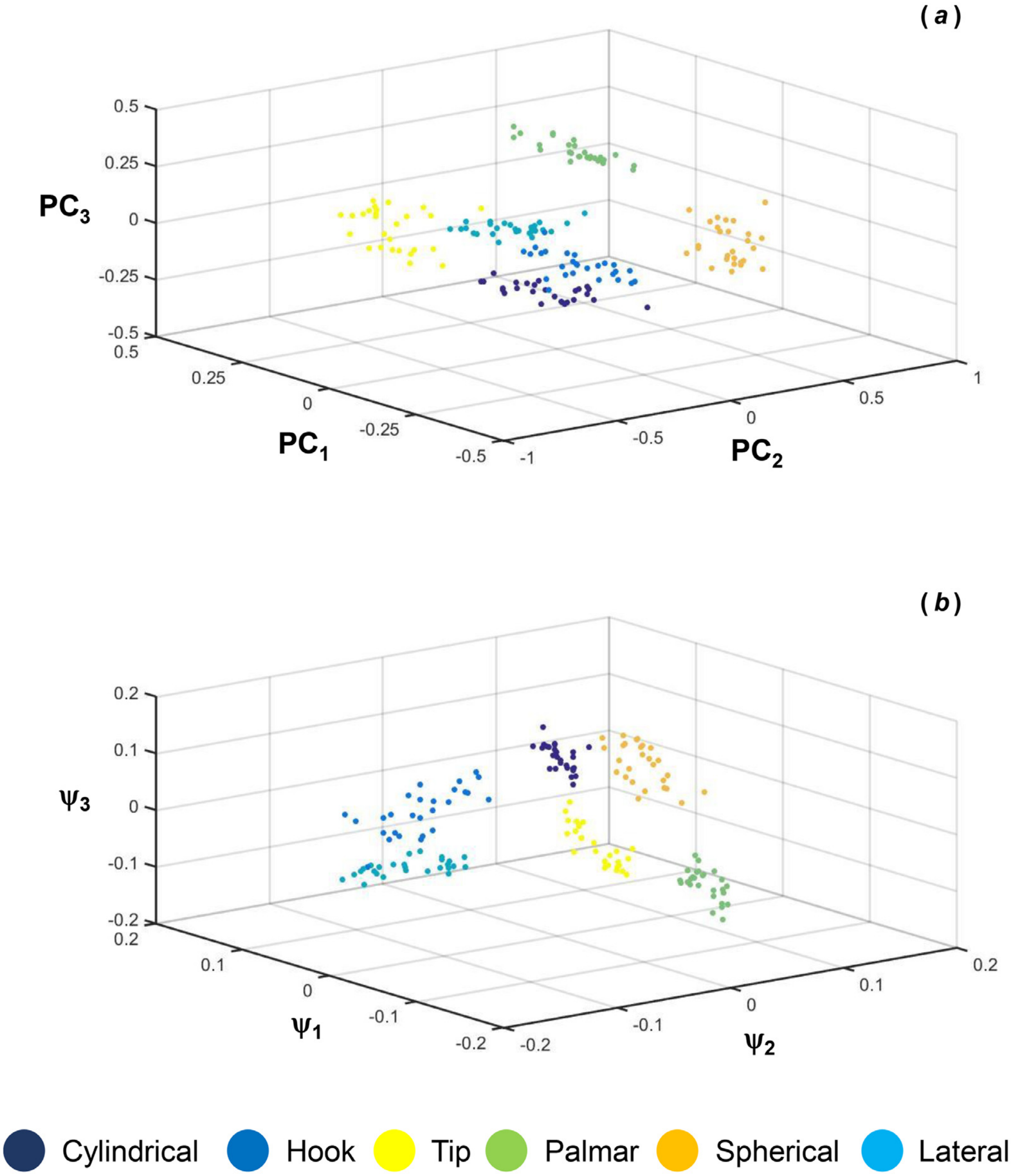

Fig. 2. An Example of 3D scatter of (a) the STFT feature after PCA (b) the STFT feature after DM of the 6 hand movements calculated from the dataset of subject # 5.

for the model construction. This study proposes a different approach, which is also suited for training data of limited size, and is easy to implement (when compared to deep learning), for constructing unified, multi-subject models. Implementation of the proposed model is done by solving a least square optimization problem, which is a common computational task in many numerical analysis and machine learning algorithms.

The goal of this paper was, therefore, twofold; to compare between the performances of nonlinear dimensionality techniques to standard linear dimensionality methods for single subject EMG hand movement classification. In particular, to examine the classification rates of a PCA and a diffusion maps (DM) based algorithm as the size of the training set decreases. Second, to propose an algorithm, which utilized data alignment, for classification of one subject's hand movements based on other subjects' hand movements based on limited size of training data. This fusion procedure aimed to bypass the variance between subjects, has rarely been studied or addressed in most of the papers in this field.

2. Methods and implementation

2.1. Mathematical background

This section provides the essential mathematical background for the dimensionality reduction and data alignment techniques, which were utilized in this work.

2.1.1. Dimension reduction methods

Let $X = \{x_i\}_{i=1}^{n}$, $x_i \in \mathbb{R}^D$ be a set of high-dimensional data points. The goal is to reduce the dimension and the complexity of the dataset while preserving some of the properties of the original dataset. In what follows, PCA (Pearson, 1901), which is a linear dimensionality reduction method and DM (Coifman & Lafon, 2006a), a non-linear dimensionality reduction method are reviewed.

The PCA method is a linear technique that projects the data into a lower dimensional space while preserving the variance. The axes of the original data space are rotated such that the new coordinate system point into the directions of highest variance of the data.

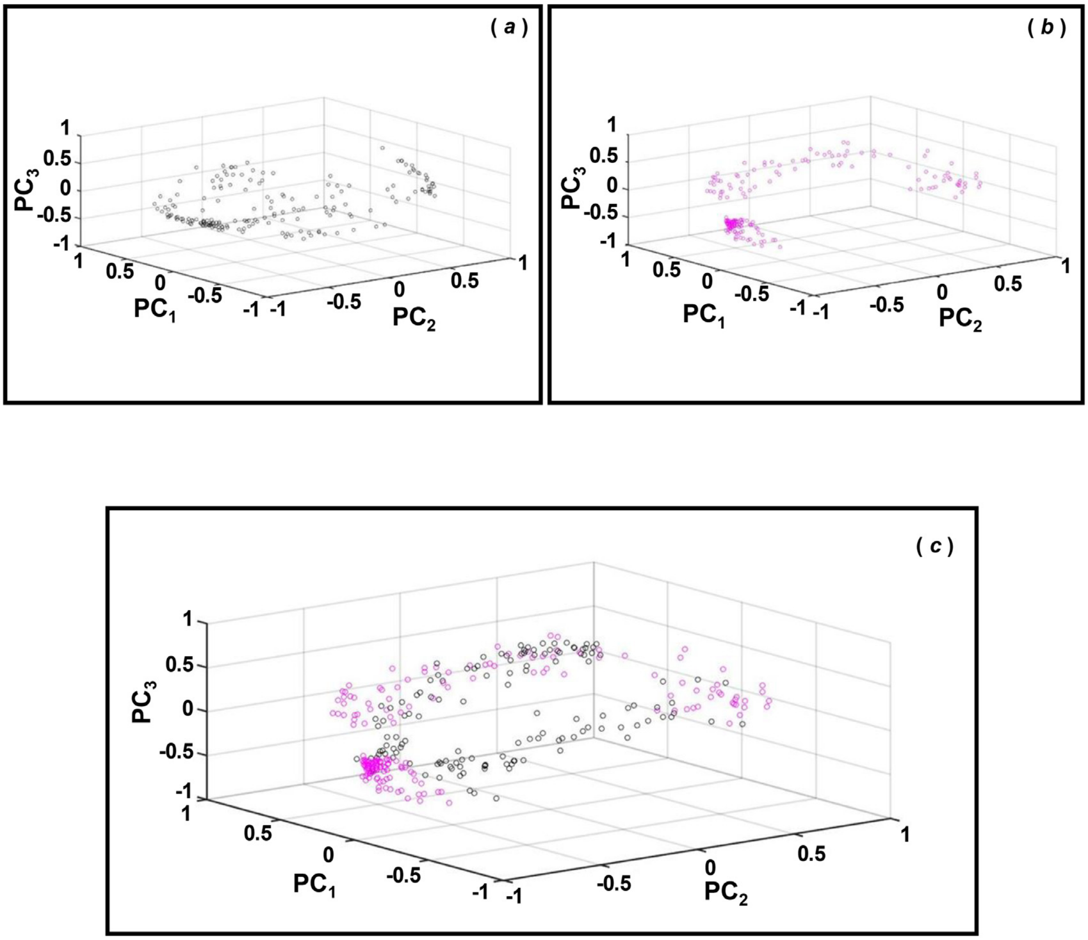

Fig. 3. Alignment of a data-set of one subject to the data-set of another subject. (a) the data set of subject # 5, (b) the dataset of subject # 4 and (c) alignment of the data set of subject # 5 on that of subject # 4.

The axes or new variables are termed principal components (PCs) and are ordered according to their contribution in describing the data in terms of minimal reconstruction error or maximal variance.

The algorithm, for the PCA method include first normalization of the dataset X to have zero mean in each attribute. Next, the covariance matrix of the data, denoted by $\Sigma_{D\times D}$is constructed and its spectral decomposition is computed. This step results in a set of D eigenvalues $\{\mu_j\}_{j=1}^{D}$ and their associated eigenvectors $\{\varphi_j(x)\}_{j=1}^{D}$. The eigenvectors describe a rotated version of the original coordinate system. The leading eigenvectors are in the directions of the maximal variance; they are the PCs. Last, the data is projected onto the first $d < <D$ PCs, $\{\varphi_j(x)\}_{j=1}^{d}$, and the dimension is reduced. The new set is denoted by $\widehat{X} = \{\widehat{x}_i\}_{i=1}^{n}$, where $\widehat{x}_i \in \mathbb{R}^d$.

In order to reduce the dimension of new data points $x^* \in \mathbb{R}^D$, the points are first normalized by subtracting the means that were calculated from the original data. Then, the points are projected onto the d PCs which yields the reduced dimension of the new points $\widehat{x}^* \in \mathbb{R}^d$.

The advantage of the PCA is that it is simple and the extension to new points is done straight forward, by projecting them into the low-dimensional space. The limitation of linear methods is that when the data has in it non-linear connections, then projecting it onto a linear subspace may result in an unfaithful model. In addition, the PCA projection is not distance preserving, meaning that two points that were close to each other in the original high-dimensional space, may be far apart in the projected space, and vice versa.

The DM method reduces the dimension of a high-dimensional dataset while preserving the local and global geometry. The associated diffusion distances define a metric on this space. The connections between the data points in the set X are studied by constructing a graph $G = (X, W)$. The graph's nodes are the data points in X. The edge weights of this graph are stored in the kernel W, which is of size $N \times N$. The chosen kernel W should be symmetric, positive preserving and positive semi-definite. The Gaussian kernel $W = w(x_i, x_j) = e^{\frac{-\|x_i - x_j\|^2}{2\sigma}}$ is a common choice for a weight function that satisfies the above properties. The kernel's scale parameter σ can be set as described in Rabin et al. (2016). Next, the kernel W is normalized. A first normalization controls the density's influence (Eq. (1)).

$$A = a(x_i, x_j) = \frac{w(x_i, x_j)}{q(x_i)q(x_j)}, \quad \text{where} \quad q(x_i) = \sum_{x_l \in X} w(x_i, x_l) \tag{1}$$

Then, a second normalization of the matrix A yields a Markov transition matrix (Eq. (2)).

$$P = p(x_i, x_j) = \frac{a(x_i, x_j)}{d(x_i)}, \quad \text{where} \quad d(x_i) = \sum_{x_l \in X} a(x_i, x_l) \tag{2}$$

The Markov matrix P (Eq. (2)) holds the probability to move from the point x_i to the point x_j in a single time step. Finally, the DMs are constructed using the spectral decomposition of P. The spectral decomposition is given by Eq. (3).

$$p(x_i, x_j) = \sum_{k \ge 0} \lambda_k \psi_k(x_i) \varphi_k(x_j) \tag{3}$$

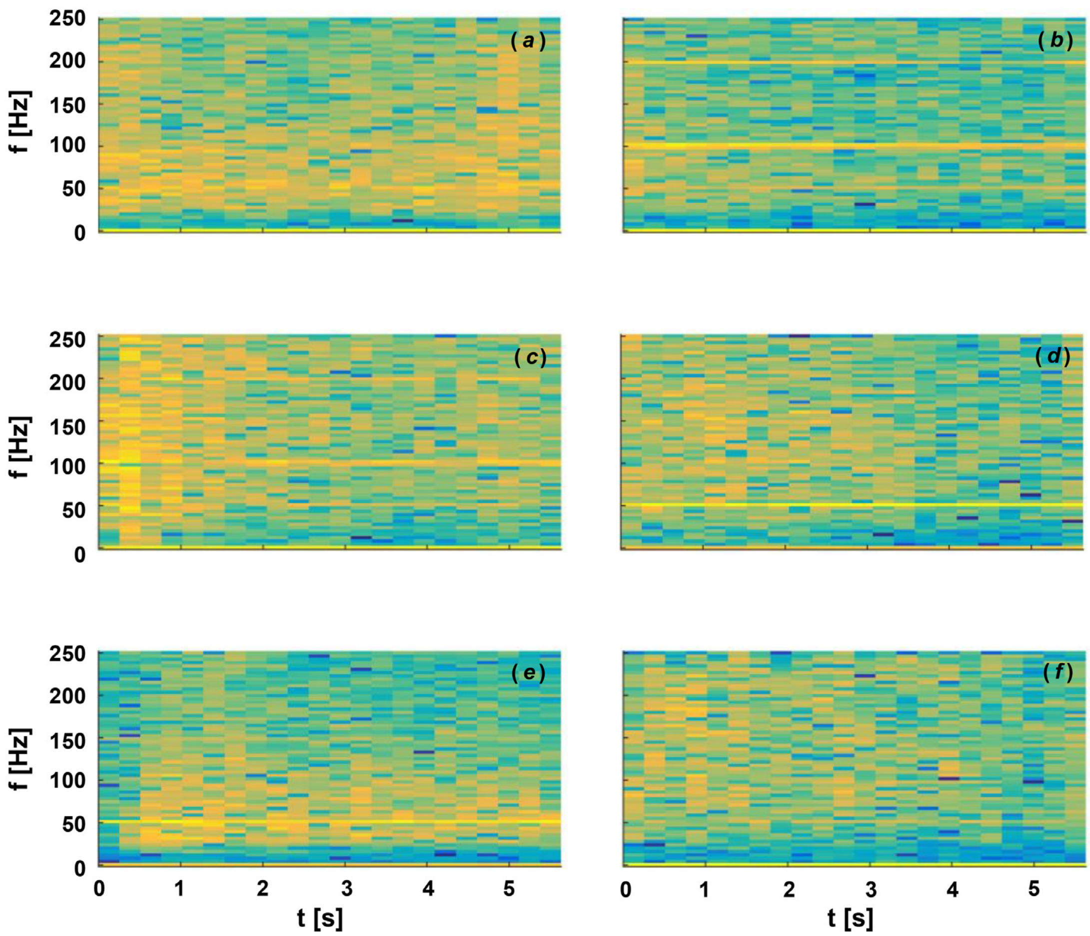

Fig. 4. An example of the spectrograms of the EMG signals of the six hand motions. a) Cylindrical, b) Hook, c) Tip, d) Palmar, e) Spherical and f) Lateral.

where $\{\psi_k\}_{k=0}^{N-1}$ and $\{\phi_k\}_{k=0}^{N-1}$ are the corresponding left and right eigenvectors of the matrix P. The properties of the matrix P yield a spectrum $\{\lambda_k\}_{k=0}^{N-1}$ that decays to zero as $k \to \infty$. Based on this property, the sum in Eq. (3) can be computed quite accurately by using only a small number of leading terms. Finally, DMs are defined by Eq. (4). The first $d < <N$coordinates from Eq. 4 are used for embedding.

$$\Psi(x_i) = \{\lambda_1\psi_1(x_i), \lambda_2\psi_2(x_i), \lambda_3\psi_3(x_i), \ldots\}. \tag{4}$$

The associated metric named diffusion distance employs the transition matrix P and its spectral decomposition to define the pairwise preserving distance metric. The diffusion distance is defined according to Eq. (5)

$$D^2\left(x_i, x_j\right) = \sum_{x_l \in X} \frac{\left(p(x_i, x_l) - p\left(x_l, x_j\right)\right)^2}{\phi_o(x_l)}, \tag{5}$$

where the denominator holds each point's density. When a large number of paths in the Markov matrix P connect between the points x_i and x_j, the diffusion distance is small. Substituting Eq. (3) into the diffusion distance definition in Eq. (5) yields the following description of the distance in terms of the eigenvectors and eigenvalues of the Markov matrix P (Eq. (6)).

$$D^2(x_i, x_j) = \sum_{x_l \in X} \lambda_k^2 (\psi(x_i) - \psi(x_j))^2. \tag{6}$$

Therefore, the DMs embedding that consist of the eigenvalues and the eigenvectors of P, preserve the geometry of the high-dimensional dataset. Since the diffusion distance between the data points as defined in Eq. (5) corresponds to the Euclidean distance between two points in the low dimensional space, then points that were close to each other in the original high-dimensional space will stay close in the embedding space. Extending DMs to include a new point $x^* \in \mathbb{R}^D$ can be achieved by utilizing the geometric harmonics method (Coifman & Lafon, 2006b), which is reviewed in the appendix.

2.1.2. Data alignment

The following data alignment technique was used for aligning between the datasets of two different subjects, after reducing the dimension of the data. Due to the variability between the subjects, the embedding of two different subjects is usually not aligned. Thus, an alignment algorithm that was introduced in (Lafon, Keller, & Coifman, 2006) for fusing embedded information from two subjects was applied.

The algorithm calculates an affine transformation to align the two embedded datasets based on a subset of matching points that are referred to as landmarks. The affine transformation is calculated in the d-dimensional embedding space. It is also possible to calculate the transformation based on a larger number of embedding coordinates $\tilde{d} > d$ for gaining more degrees of freedom. Then, the calculated rotation is projected onto the desired number of coordinates, d.

Let $X = \{x_1, x_2, \ldots x_{N_1}\}$ and $Y = \{y_1, y_2, \ldots y_{N_2}\}$ be two datasets of size N_1 and N_2, respectively. The points $\{x_i\}_{i=1}^{N_1}$ and $\{y_i\}_{i=1}^{N_2}$ are embedded in a low-dimensional space. Assume that the first K points in each set $\{x_1,x_2,\ldots x_K\}$, $\{y_1,y_2,\ldots y_K\}$ are the pairs of landmarks for the alignment algorithm. The affine function $g : \mathbb{R}^{K-1} \to$

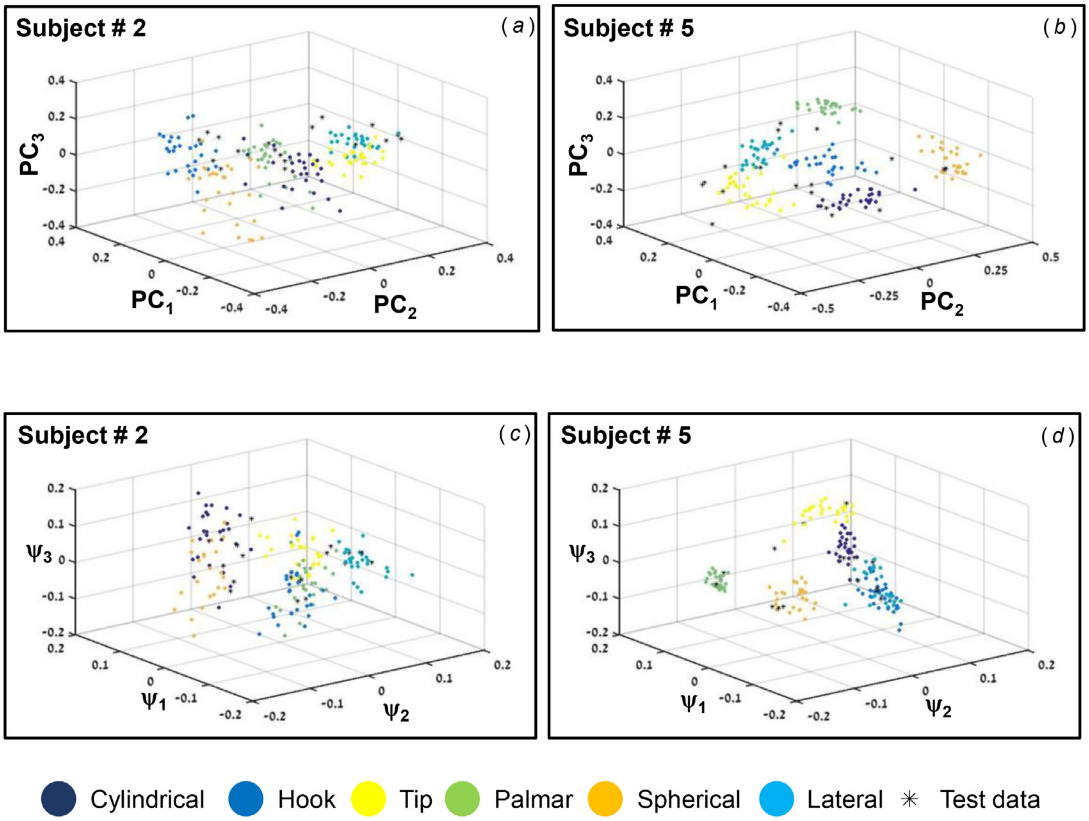

Fig. 5. An Example of 3D scatter of the 6 hand movements, calculated from the dataset of subject # 2 and # 5, of the STFT feature after PCA (a and b) and of STFT feature after DM (c andd). The circles represent the training data (90% of the dataset) while the stars represent the test data (10% of the dataset).

Table 1
The average success rates of single subject classification of all subjects at each size of the training set.

Train size [%]	Average success rates [% ± SD]	
	PCA	DM
90	94.8 ± 3	88 ± 10.3
50	91.6 ± 3.2	85.7 ± 10.8
33	84 ± 8	84.8 ± 12.6
17	69 ± 6	81.5 ± 12.5

$\mathbb{R}^{K-1}$ that satisfies $g(x_i) = y_i$ is estimated by computing $h(x_i) = \arg\min_{y \in Y}\{\|g(x_i) - y\|\}$.

The affine transformation, defined by $g(x_i) = Mx_i + b$ is composed of a linear transformation M of the space X and a vector b. The matrix M is of size $\tilde{d} \times \tilde{d}$ and the vector b is of size $\tilde{d} \times 1$. The minimum is solved by using the total least squares approach, This computation includes matrix multiplications and matrix inversion. The Cholesky decomposition (Gentle, 1998) may be used to accelerate the matrix inversion parts. Then, the computation complexity is at most $O(\tilde{d}^2K)$. Then, the alignment function is used to align the rest of the points in X to Y.

2.2. Dataset

Raw data of EMG measurements were taken from the public UCI repository (UCI datasets). This dataset was previously used by Sapsanis et al. (2013). The data was measured from five subjects while performing the following six movements (Fig. 1): a) holding cylindrical tools (Cylindrical), b) supporting a heavy load (Hook), c) holding small tools (Tip), d) grasping with palm facing the object (Palmar), e) holding spherical tools (Spherical) and f) holding thin objects (Lateral).

Each movement was recorded by two channels that measured the electrical activity of the hand muscles. Subjects were asked to repeat each movement for 30 times. The recordings included only the records of the muscle activity, meaning there were no need for segmentation.

2.3. Feature extraction

The features extraction step was carried out by using the STFT. Each waveform was transformed to be represented by a two-dimensional time-frequency matrix. The time window size was set to be 256 and with an overlap of 128 between them.

Since each movement was recorded by two channels, two corresponding spectrograms were calculated and then concatenated to form one united feature matrix that represents a single movement.

2.4. Classification of movement for each subject based on its own data set

The classification of movement for each subject was done based on a subset of known, classified movements that the subject performed. Two algorithms for single subject movement classification were used. The two algorithms differ by the methods used for dimensionality reduction. Both of the algorithms utilized the STFT feature matrix that was constructed for each subject . This matrix includes 2860 features. Thus, the linear PCA or the non-linear DM methods were applied to reduce the dimension of the feature space to 15. A 3D scatter of the STFT features set for all 6 hand movements after applying the PCA and DM methods are represented in Fig. 2. Each color represents a specific movement and each point represents a sample in the features data set.

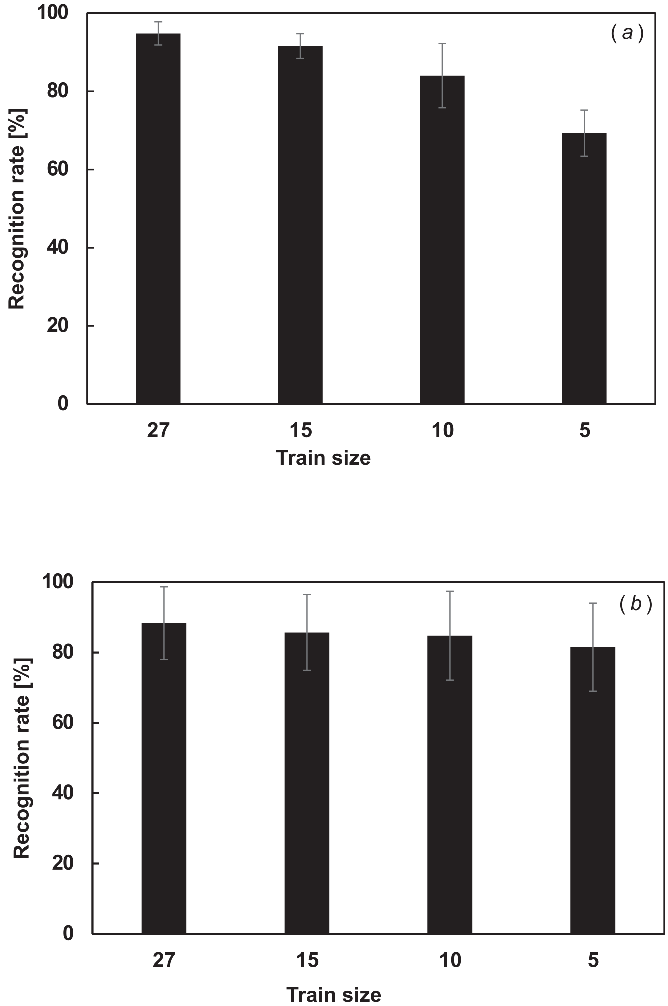

Fig. 6. The average recognition success rates of the 6 hand movements of all subjects (mean±SD) with the decrease in the train size. Movement classification rates were calculated for each subject based on its own data set. a) the STFT features after PCA, b) the STFT features after DM.

KNN with 3 neighbors and 10 folds of cross-validation was used to classify new movements. The performance of the two algorithms (PCA and DM) was evaluated by a train phase, in which the low-dimensional representation of the known movements was constructed, followed by a classifications phase that processes new, unlabeled movements and returns a prediction of their type. Four train set sizes, corresponding to 90%, 50%, 33% and 17% of the data set, were analyzed. For each train set size, 10 classification round were performed. At each round different samples from each movement in the movement set were randomly chosen for the test dataset and the remaining samples were a part of the training data-set. For each train set size the average classification rates (±SD) were calculated.

2.5. Classification of movement for each subject based on another subject data set

The affine transformation method described in Section 2.1.2 was used to classify movements of a given subject based on a small number of labeled movements that the subject performed and on a labeled training dataset of movements that belong to other train subject. The dataset that was used in this work contained data from 5 subjects, the classification of new movements for a given subject was based on the labeled datasets of 3 arbitrary training subjects.

The algorithm for this part included training phase, affine transformation and classification of a test movement. In the training phase STFT features were extracted for all of the movements that belong to the selected training subjects and to a small subset of labels data points that belong to the test subject. Here, 5 labeled movements of each type are assumed to be known in advance for the test subject. Next, the feature dataset of each subject is embedded into a low dimensional space separately using the PCA or DM methods. This resulted in 4 low-dimensional representations, 3 of them belonging to the training subject, and the forth one, which consisted of a smaller number of data points, belongs to the test subject.

The alignment algorithm was then applied to rotate and merge the low-dimensional representation of the test subject onto the low-dimensional representation of the train subject. Here, since the train data consists of 3 subject, the alignment procedure was done 3 times, each rotated the test subject's embedded data onto the embedding of a different train subject. The alignment procedure was based on 5 landmark points per movement. In favor of stabilizing the alignment process, the procedure of rotating the embedded test subject's data onto an embedded train subject's was repeated 3 times. Each time the 5 landmark point's that belong to the train subject are randomly chosen. Note that test subject has a small number of labeled data points for each movement, thus the test subject's landmarks contains this small labeled set and this set is used for all of the alignments applications.

For example, classification of the hand movements that belong to subject #1, given the full labeled datasets of subjects #2, #3, and #4 was done in the following manner:

Subject #1 has 5 labeled movements of each type which resulted in a total of 30 labeled feature vectors. The training set subject each hold 180 labeled feature vectors (30 for each movement). Each subject's feature space was independently embedded into a low-dimensional space. Then, the embedding of test subject #1 was aligned to the other three embeddings by randomly choosing 30 landmark points from each of train subject's embeddings. An example of the low-dimensional representations belonging to the train and test subjects before and after the alignment of the test subject on the train subject is seen in Fig. 3.

Finally, classification of a test movement (that belongs to subject #1) was computed as follows: STFT features were extracted from the new movement and the feature vector was embedded in the low-dimensional (PCA or DM) space that belongs to the test subject. Then, the computed affined transformations was applied to new embedded test points, this rotated the test point into the train subject's embeddings. The test subject was rotated to each of 3 train subjects 3 times, thus this alignment step resulted in 9 rotations, three different rotations per train subject. KNN was then used to classify the new embedded test point in each of the 9 low-dimensional representations. The final classification result was determined by a double majority vote. First, a majority classification vote was calculated with respect to the alignments of each train subject separately. This step results in 3 classifications, one based of the alignment to each train subject. Then, the majority from the 3 votes was chosen to be the predicted class label for the new test subject's movement. The average (±SD) classification rates of all subjects while using the PCA or DM methods were then calculated.

3. Results

3.1. STFT

All EMG signals that were acquired during 6 hand motions, were padded with zeros according to the signal with the maximal

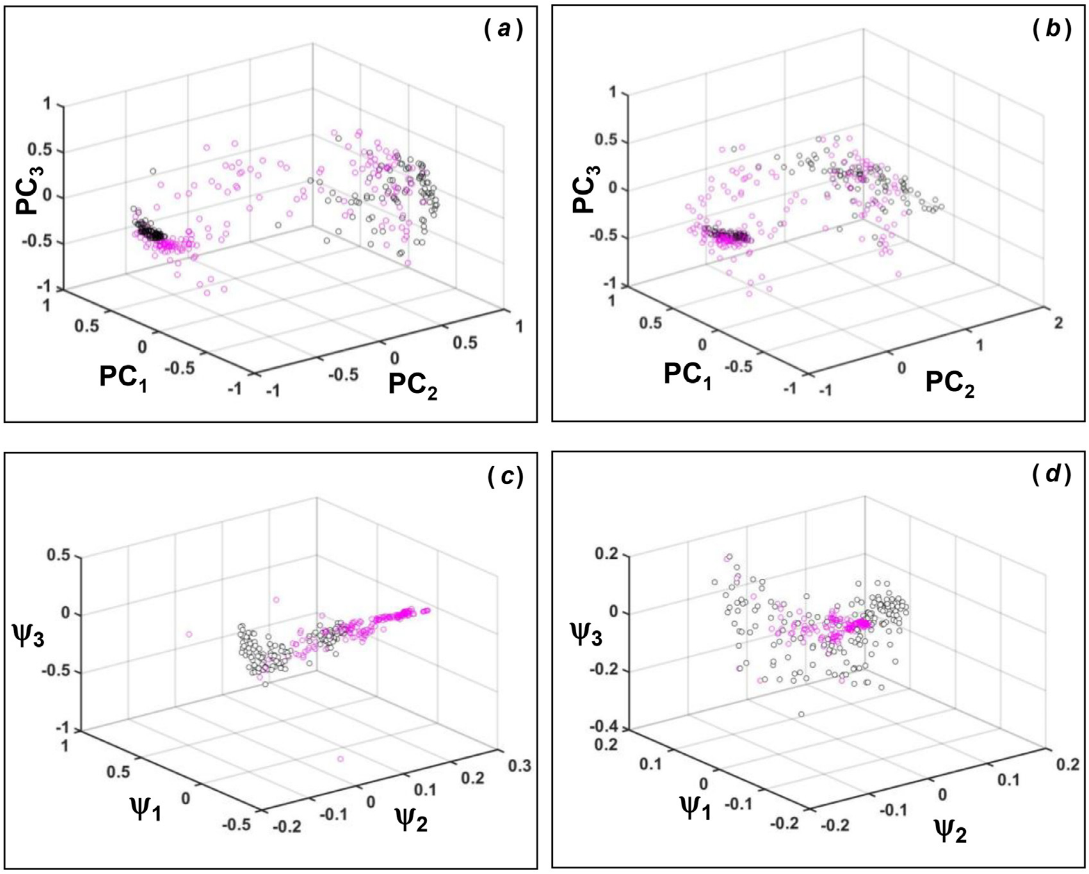

Fig. 7. 3D scatter of STFT feature with PCA and with DM of subject #1 (purple circles) and subject # 2 (black circles). a) 3D scatter of STFT feature with PCA of both subjects before the alignment, b) 3D scatter of STFT feature with PCA after the alignment of the data of subject # 1 on that of subject # 2, c) 3D scatter of STFT feature with DM of both subjects before the alignment d) 3D scatter of STFT feature with DM after the alignment of the data of subject # 2 on that of subject # 1.

length in the dataset. STFT algorithm was then applied to all EMG signals using a window size of 256 with an overlap of 50% to yield the changes in the frequency distribution of the signals over time. An example of the STFT of 6 EMG signals, each represents one of the 6 hand motions is described in Fig. 4. The colors in the Fig represent the intensity (amplitude) of the signal at each frequency. As can be seen from the representative figure different hand movements reveal EMG signals with different frequency distribution and intensity.

3.2. Classification of movement for each subject based on its own data set

To examine the ability to classify between the different motions for each subject KNN with 3 neighbors was applied using 10 folds of cross-validation. The KNN algorithm was applied on the STFT features after applying the PCA and on the STFT features after applying the DM. To evaluate the size of the train set needed in order to achieve relative high recognition rates four train set sizes were analyzed: 27, 15, 10 and 5 repetitions. This train set sizes correspond to 90%, 50%, 33% and 17% of the data set. Fig. 5 described 3D distribution of the STFT features with PCA and of the STFT features with DM of 2 subjects. The colors points represents the train set (90% of the dataset) and the black stars the test set. Each color of the colors points in the figure represents different hand motion. A relative good separation between the motions is seen in the distribution of the STFT features after applying both the PCA (Fig. 5a and b) the DM (Fig. 5c and d) for both the training and the test sets for each subjects.

The recognition success rates for the STFT features with PCA and for the STFT features with DM decrease with the decrease in the size of the training set (Fig. 6). The average success rates of all subjects at each size of the training set for both PCA and DM methods are summarized in Table 1. The average success rates decrease from 94.8 $\pm$ 3% and 88.3 $\pm$ 10% at train set size of 90% to 69.3 $\pm$ 6% and 81.3 $\pm$ 12% at train set size of 17% while using the STFT features with PCA and for the STFT features with DM, correspondingly.

3.3. Classification of movement for each subject based on another subject data set

In order to classify hand movement of one subject based on a data-set constructed from signals of another subject data alignment technique was used. As described in the method section, each test data-set was aligned to the training data-set using an affine transform. For each subject, the embedding of its test data was aligned to the data of the other trained subjects by randomly choosing 30 landmark points from the train data. In this work, the procedure was repeated for robustness and the data of the test subject was aligned to each of the data of the train subject 3 times. Fig. 7 shows the data set of two subjects before and after the alignment of the test subject on the trained subject. Both data-sets were based on STFT features after dimensionality reduction was applied using either the PCA method (Fig. 7a and b) or the DM method (Fig. 7c and d). In both dimensionality reduction methods the alignment algorithm reduce the variability between the data points of the two subjects and more data from the

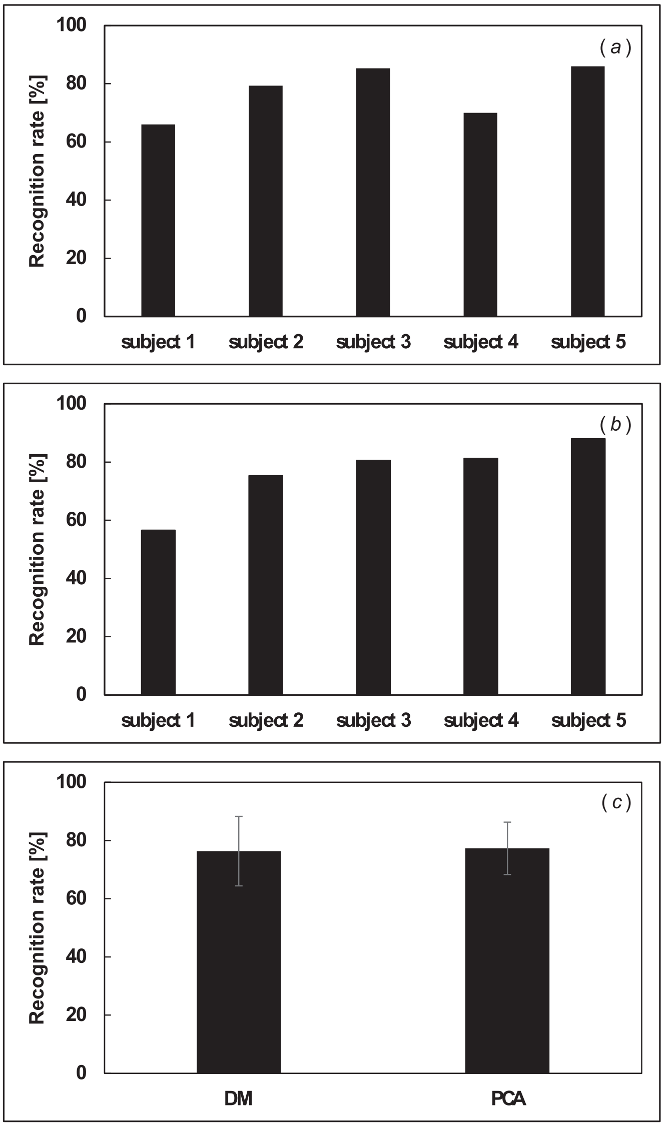

Fig. 8. The average recognition success rates of the 6 hand movements. Movement classification was done for Each Subject based on another subject data set a) the STFT features after PCA, b) the STFT features after DM, c) the average classification rates (mean±SD) calculated from all 5 subjects.

tested subject converged with those of the trained subject (Fig. 7b and d).

For each subject KNN classifier was used to classify the new embedded test point, a majority classification vote was calculated with respect to the alignments of each train subject separately. The final classification result was determined by a second majority vote from the 3 classifications, each one based of the alignment to each train subject. Classification success rates while using the PCA method were 66%, 79%, 85%, 70% and 86% (Fig. 8a) and 56%, 75%, 81%, 81% and 88% (Fig. 8b) while using the DM method for subject # 1–5, respectively (Table 2). There was no significant difference between the average success rates of all subjects (Fig. 8c) while using the PCA method (77 ± 9%) compare to that achieve while using the DM method (76 ± 12%).

Table 2
The average success rates for each subject based on another subject data set.

	Average success rates [% ± SD]	
	PCA	DM
Subject 1	66	57
Subject 2	79	75
Subject 3	85	81
Subject 4	70	81
Subject 5	86	88
Average all	77.3 ± 9	76.4 ± 12

4. Discussion and conclusion

The research aimed to compare between linear and non-linear dimensionality reduction techniques and to introduce an alignment algorithm for fusion multi-subject data set. These methods were implemented for classification of human hand movements based on EMG signals. A data set of EMG signals recorded from five healthy subjects was taken from an open public source. EMG features were extracted by application of the STFT. PCA and DM methods were used for dimension reduction and their performances were compared in the setting of single subject and multi subjects. Classification was done by utilizing the KNN algorithm on the reduced spaces.

Selection of features for characterizing EMG signals have been the subject of many studies over the last decades (Al-Angari et al., 2016; Khushaba et al., 2016; Phinyomark et al., 2013; Tsai et al., 2015). Here the STFT, which capture information both in the time and frequency domain was selected for the feature extraction step. This feature was found previously to be significant and informative when compared to conventional EMG features (Tsai et al., 2015).

An important property of classification methods is their ability to identify pattern based on small or limited dataset especially for application that required personal customization. In this study different sizes of training sets, ranging from 90%, 50%, 33% to 17% were examined. Although, there was a large reduction in the size of the training set (from 90% to 17%) the decrease in the classification results were relatively moderate. When comparing between the two dimension reduction methods, while PCA provided higher results for the case of the larger training set (94.8 ± 3% vs. 88.3 ± 10% at train set size of 90%) the DM method was proved to be more stable for classification based on a smaller train set (69.3%±6 vs. 81.3 ± 12% at train set size of 17%). The results for large training set obtained in both methods are comparable with previous studies that achieved accuracy rates of 85%, 89% and 93% (Chu et al., 2006; Mane et al., 2015; Sapsanis et al., 2013; Tang et al., 2012). It should be noted that when comparing the obtained results to other studies that used the same dataset, the train-test data separation should be taken into account. The single subject classification that was perfumed in Sapsanis et al. (2013) and in Gu et al. (2017) is similar to the way single subject classification was carried out in this work. Their correct average classification rate in (Sapnsnais et al.) was 85% and 91%, correspondingly. These results are comparable with our PCA and DM based single subject classification algorithm, which result in accuracies of 94.8% and 88.3%, respectively. Ruangpaisarn and Jaiyen (2015) and Ramírez-Martínez et al. (2019) achieved higher classification rate than the results obtained in this study, however, a training set of larger size with a larger number of features was used. The stability of the DM method for the smaller train set shows the method's advantage for real life applications, where it is not realistic to collect a large number of training samples for identification of movements from each new subject.

A major challenge in many medical and other human related application is to handle the large variances between the signals measured from different subjects. To overcome this challenge, which is just recently being addressed by deep learning methods (Côté-Allard et al., 2019; Phinyomark & Scheme, 2018; Tsinganos et al., 2018), a new approach for aligning between the

Table 3
Summary of studies conducted for EMG based hand movement classifications.

Authors	Average accuracy rates	Methods
Chu et al. (2006) [3]	95.5% 97.5%	Wavelet transform PCA A self-organizing feature map Multilayer perceptron Single subject classification
Tang et al. (2012) [30]	89%	Energy ration and concordance correlation features and LDA classifier Single subject classification
Sapsanis et al. (2013) [28]	85%	PCA and Relief algorithm Single subject classification
Isaković et al. (2014) [14]	77.2% 64.5% 46.6%	PCA A piecewise quadratic classifier Classifying set of grasping movements Classifying set of wrist movements Classifying set of eleven finger movements Single subject classification
Mane et al. (2015) [18]	93%	Wavelet transform and ANN
Ruangpaisarn and Jaiyen (2015) [27]	91.66% 94% 94.77% 98.2%	Singular value decomposition Naïve Bayes Radial Basis Funcion Netwirk KNN SVM Single subject classification
Al-Angari et al. (2016) [1]	94%	Correlation-based method and Distance-based method SVM Single subject classification
Gu et al. (2017) [13]	85% 55% 65% 91% and 94%	EMD features Neural Network Adaptive Boosing Linear discriminant Analysis Random Forest without and with PCA Single subject classification
Ramírez-Martínez et al. (2019) [26]	83% 99.8% 99.2% 97.5% 80.4%	Time domain and autoregressive model features. Bayes net Instance-based classifier Multilayer perceptron Decision trees SVM Single subject classification
Tsinganos et al. (2018) [32]	70.48% and 72%	Deep learning based on convolutional neural networks Multi subject classification
Côté-Allard et al., 2019 [6]	68.98% 98.31%	Deep learning Raw EMG input Continuous wavelet transform input Multi subject classification
This Study	94.8% 88% 77.3% 76.4%	Single subject classification: PCA DM Multi subject classification: PCA DM

reduced feature spaces of two subjects was proposed in this study. In this proposed algorithm the train subjects had a full, large training set, which included many examples of each hand movement type, while the test subject had only limited amount of train data. Since the train subjects had a full feature space, it allows to compute several different alignment of the limited feature space based on randomly choosing landmark points. For each different pairs of train-test alignments, multiple reduced feature space models were obtained, thus, enables a more robust classification by using majority votes. The robustness of the methods is clearly reflected in the classification results where success rates of 77 ± 9% and 79 ± 12% was found for the PCA and DM method, correspondingly. Recall, that for this small 17% training data set size, the result of a single subject classification algorithm was relatively lower (69.3% ± 6) for the PCA and quite similar (81.3 ± 12) for DM. It can be assumed that using a larger group of labeled train subjects would have increased the classification success rates for any new subject. Hence, when performing EMG based movement identification of a new subject, this alignment method provides a robust solution, which requires only a small number of measurements form the new subject. The stability, found here, of the DM method for the smaller train set both for a single subject and multi subject classification shows the method's advantage. Creating reliable algorithms for limited size data has an important implications for real life applications, where it is not realistic to collect a large number of training samples for identification of movements from each new subject. Additionally, when compared to deep learning solutions for merging datasets of different subjects, our proposed approach benefits from a simple implementation, which only requires to compute a least squares optimization.

Although the results of the present project show high accuracy rate for a single subject classification and relative good rate for multi subjects classification compared to other studies conducted for EMG based hand movements classification (Table 3), several issue must be considered. First, the data set was taken from an open public source. Therefore, we didn't have enough information on the position of the electrodes, possible artifacts, the quality of the record and we could not confirm that the EMG data were indeed correspond to a specific movement etc. In order to establish our findings and to control the experimental protocol and dataset more measurements should be done to create a database of signals from various subjects and additional hand movements. Furthermore, the influence of using other features and other classifiers should also be examined, especially when using the alignment method.

The main findings of this study show that STFT can be used as a single informative feature for movement identification, strengthens previous studies that suggest using STFT for feature extraction. Both the DM and PCA give high classification rates in a single subject setting when using a large training data set. In case of limited training data the DM outperformed the PCA. This suggest that nonlinear dimensionality reduction techniques is more reliable and less sensitive to amount of sampled in the input training set. Additionally, the novel classification approach, aimed to reduce the large variance between different subjects by utilizing dimension reduction methods along with an alignment algorithm was proved to enable robust classification of a new subject based on other subjects' data sets. Furthermore, the obtained results also suggest that data alignment performed in a reliable low-dimensional space can successfully merge limited samples from the tested subject with a richer training set of other subjects. Since the alignment is performed in a low-dimensional space and is based on a limited number of anchor points, the computation complexity is not high. Last, this proposed algorithm framework is general and may be applied to other biomedical classification tasks in which large variability exists between the measured datasets that are recorded from different subjects. These applications include classification of other limbs movements, speech recognition based on EMG from facial muscles, identification of muscle fatigue and identification of respiratory weakness and airway obstructions based on EMG signals from respiratory muscles.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The geometric harmonics (GH) (Coifman & Lafon, 2006b) is a method for data extension. The main idea is to construct a basis from a set of harmonic functions. The basis can describe any general function, and in particular is can span the diffusion maps coordinates. Like in the diffusion maps method, the construction of geometric harmonics includes the construction of a kernel, which is positive and semi-definite, thus its spectrum decays. This property allows to use a small number of elements in the basis.

Let $X = \{x_i\}_{i=1}^{n}, \quad x_i \in \mathbb{R}^D$ be a set of high-dimensional points and f is a function that is defined on the data, which we wish to extend to new data points. In the context of this paper, the functions to be extended are the diffusion maps coordinates. Assume for simplicity that $f(x_i) = \lambda_1 \psi_1(x_i)$. Let $\bar{x} \in \mathbb{R}^D$ be a new point, the goal is to evaluate $f(\bar{x}) = \lambda_1 \psi_1(\bar{x})$.

First, a Gaussian kernel is constructed on the dataset. The kernel width σ is the extension scale of the kernel, and it should be set to be larger than the scale σ, which is used in DM. The extension scale σ depends on the smoothness of the function f Next, the spectral decomposition of the kernel is computed by

$$\mu_l \upsilon_l(x_i) = \sum_{x_j \in X} e^{\frac{-\|x_i - x_j\|^2}{2\sigma}} \upsilon_l(x_j),$$

where $\{\mu_l\}_{l=0}^{n-1}$, $\{\upsilon_l\}_{l=0}^{n-1}$ are the eigenvalues and corresponding eigenvectors of the kernel.

The function f may be written as a linear combination of this basis

$$f(x_i) = \sum_l \langle f, \upsilon_l \rangle \upsilon_l(x_j), \quad x_j \in X.$$

The eigenvectors can be extended to the new data point $\bar{x} \in \mathbb{R}^D$, assuming that $\mu_l \neq 0$, thus

$$\bar{\upsilon}_l(\bar{x}) = \frac{1}{\mu_l} \sum_{x_j \in X} e^{\frac{-\|\bar{x} - x_j\|^2}{2\sigma}} \upsilon_l(x_j).$$

The extended eigenvectors are used to evaluate the function $f(\bar{x})$ by

$$f(\bar{x}) = \sum_l \langle f, \bar{\upsilon}_l \rangle \bar{\upsilon}_l(x_j), \quad x_j \in X.$$

Since the spectrum decays, the last sum should contain only a small number of elements. In GH, a condition number δneeds to be determined and in practice the last sum should only contain elements that satisfy $\mu_l \geq \delta \mu_0$.

Credit authorship contribution statement

Neta Rabin: Conceptualization, Methodology, Writing - original draft. **Maayan Kahlon:** Data curation. **Sarit Malayev:** Data curation. **Anat Ratnovsky:** Conceptualization, Methodology, Writing - original draft.

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